

Akshaj Gandhi

☎ +91 9573276206 ✉ akshajsaigandi@gmail.com 🔗 LinkedIn 🌐 GitHub 📍 Hyderabad, Telangana

Professional Summary

Electronics and Communication Engineering undergraduate (CGPA: 9.67/10) specializing in RTL Design, FPGA Development, Computer Architecture and RISC-V Processor Design. Designed Verilog-based hardware systems, published an Indian patent application, secured Top 3 in the Currents 2026 Hackathon (NIT Tiruchirappalli), and seeking RTL/FPGA/VLSI internship opportunities.

Education

SRM Institute of Science and Technology, Tiruchirappalli 2024 – 2028 (Expected)
Bachelor of Technology in Electronics and Communication Engineering
CGPA: **9.67/10.00** (As of Semester-IV)

FIITJEE Junior College, Hyderabad 2022 – 2024
Intermediate (Class XI – XII)
Percentage: **92.6%**

Technical Skills

Programming Languages: C, Python

Hardware Description Languages: Verilog HDL, VHDL, SystemVerilog

Hardware Design: RTL Design, FPGA Design, Digital Logic Design, Computer Organization & Architecture, RISC-V Processor Design

EDA & Development Tools: Xilinx Vivado, MATLAB, Git, GitHub

Operating Systems: Windows, Linux

Relevant Coursework

Computer Organization & Architecture • Digital Logic Design • Advanced Digital System Design • Solid State Devices • Signal Processing • Electronic System & PCB Design • Programming for Problem Solving

Projects

RVV-Inspired RISC-V Vector Processing Unit

Verilog HDL — Xilinx Vivado

GitHub

- Designed and implemented an RV32I processor with an RVV-inspired vector extension using Verilog HDL.
- Developed scalar datapath modules, vector register file, vector ALU and vector control unit for parallel vector execution.
- Verified RTL functionality using simulation and integrated scalar and vector execution into a unified processor architecture.

Optimized Pipelined Carry-Save Adder (PCSA)

Verilog HDL — Xilinx Vivado — FPGA

GitHub

- Designed a 16-bit pipelined Carry-Save Adder for high-speed cryptographic arithmetic.
- Implemented RTL in Verilog HDL and synthesized the design on a Xilinx Artix-7 FPGA.
- Improved throughput through pipelining while evaluating timing performance and FPGA resource utilization.

Neuromorphic VLSI Architecture for On-Chip Spiking Neural Network Acceleration

Python

GitHub

- Developed a Leaky Integrate-and-Fire (LIF) neuron-based neuromorphic architecture.
- Simulated spiking neural network behaviour using Python for hardware-oriented neural computation.
- Documented architecture, mathematical model and simulation workflow for reproducible evaluation.

Achievements

Secured Top 3 in Currents 2026 Hackathon, NIT Tiruchirappalli.

Patent

Published Indian Patent Application - Optimized Pipelined Carry-Save Adder (PCSA), Application No. **202641018710**, Publication No. **IN202641018710 A1**

Certifications

VLSI Chip Design with CPS for Industrial Applications (Specialization) • Design of CPS with ARM Processor using Embedded C • CPS Design with ARM Core using MicroPython • CPS Design for Mechatronics, Healthcare, EV & Robotics • Introduction to Computer Organization • 6G Evolution: Blockchain, Semantic Communications & Radar

Languages

English • Telugu • Hindi • Tamil (Basic) • German (Basic)